

Laboratory Handout

Liquid-liquid extraction, also called solvent extraction, is a separation process in which two immiscible liquids are brought into contact in order to obtain a solute or to remove an impurity.

Distillation is generally preferred in chemical processing, but where separation by distillation is inefficient – when the components form an azeotrope, or when they are heat-sensitive – liquid-liquid extraction is considered as an alternative. Because the separation depends on chemical properties rather than on volatility, it is also well suited to high-purity and heat-sensitive products, and the growing demand for such products has made it an important part of chemical processing. A wide variety of extraction equipment has been developed to meet this demand and is in use throughout industry.

In this experiment, one such device, a packed column, is used to extract a solute present in one liquid into another liquid that is immiscible with it.

1. Objective

A liquid containing the solute (carrier) is contacted with an immiscible liquid (solvent) in the extraction column, the concentration of the extracted solute is measured, and the number of theoretical stages is determined. The number of stages is then examined under different operating conditions, so as to understand the principles of mass transfer and the factors that govern the extraction process.

2. Materials and Apparatus

Apparatus

- UOP5 Liquid-Liquid Extraction Unit (Armfield Ltd., UK), shown in Fig. 1 – a vertical glass column (50 mm inner diameter, 1160 mm packed height) filled with glass Raschig rings to a void fraction of 0.65

Chemicals

- n-Bromopropane (99.0%, Samchun Chemical, Korea) – carrier
- Propionic acid (99.0%, Samchun Chemical) – solute
- Distilled water – solvent
- Sodium hydroxide (98.0%, Samchun Chemical) – titrant
- Phenolphthalein solution (1%, Samchun Chemical) – indicator, color change over pH 7.8–10.0



Figure 1: The UOP5 Liquid-Liquid Extraction Unit (Armfield Ltd., UK).

3. Safety

Wear safety glasses, a laboratory coat, and gloves throughout. n-Bromopropane is a volatile organic solvent: handle it in a fume hood or a well-ventilated area, avoid skin contact, and avoid inhaling the vapor when samples are drawn and waste is discarded.

4. Experimental Procedure

The letters and numbers in parentheses are the tags printed on the control panel of the unit; the full control and instrumentation diagram is given in the manufacturer's manual supplied with the unit.

Preparation

1. Switch on the unit at the main switch (S1) and start the chiller.
2. Check the level in the distilled water reservoir (L2) and top it up if necessary.
3. Charge the feed, a mixture of n-bromopropane and propionic acid, to the feed tank (L5).
4. Close the column feed valve (V8), open V13, and run the feed pump (S4) to measure the feed flow rate. Stop the pump, then close V13 and open V8.
5. Prepare 500 mL of standardized 0.5 M aqueous sodium hydroxide solution.

Extraction run

6. Start the water pump (S3) and set the rotameter (F1) to its maximum to fill the column up to the packing.
7. Once the column is filled to the packing, reduce the water flow rate to 100 mL min⁻¹.
8. Start the feed pump (S4).
9. When the heavy phase collecting at the base of the column rises above the lower electrode, move S2 down so that the raffinate drains to its receiver (L3). The interface is then held at a fixed level by the electrodes.
10. Run the column for about 30 min, checking that both flow rates stay constant, until steady state is reached.

Sampling and titration

11. At steady state, collect the feed from the feed tank (V6), the extract from the water overflowing to its receiver (L1), and the raffinate from its receiver (V9).
12. Dilute the feed and the extract tenfold, 1 mL of sample with 9 mL of distilled water, to give 10 mL samples; take the raffinate undiluted as a 10 mL sample.
13. Titrate each sample with the standardized 0.5 M aqueous sodium hydroxide solution to the phenolphthalein end point.

5. Data Analysis and Report Requirements

Include the following in the report. The equilibrium and tie-line data for the n-bromopropane + propionic acid + water system are bundled with CBPL-kit.

Calculations

1. Take the density of the feed and the raffinate as that of pure n-bromopropane, 1.354 g mL^{-1} , and the density of the extract as that of pure water, 0.997 g mL^{-1} .
2. From the measured volumetric flow rates, calculate the mass flow rates of the feed and the solvent.
3. From the titre of sodium hydroxide solution for each sample, calculate the concentration of propionic acid (mol L^{-1}) and its mass fraction in the feed, the extract, and the raffinate.
4. Plot the binodal curve and the tie-lines on a ternary diagram.
5. Determine the number of theoretical stages by the Hunter–Nash construction.
6. Obtain the extract composition E'_1 expected at the experimental solvent-to-feed ratio from the lever rule, and compare it with the measured extract composition E_1 .

Discussion

1. Which phase was dispersed and which was continuous in this run? Could the two be interchanged, and what would change if they were? What role does the packing play for each phase?
2. Account for the difference between the extract composition obtained from the lever rule and the measured composition. How could the agreement be improved?
3. Using the Hunter–Nash construction, explain how the number of theoretical stages responds to a change in the solvent-to-feed ratio and to a change in the feed composition.